SYNTHETIC SAMPLE REPORT · FICTITIOUS PATIENT · NOT FOR CLINICAL USE

MULK LONGEVITY

Comprehensive Biomarker Panel

6 high | 6 low | 42 in range

PATIENT	PATIENT ID	SEX / AGE	DOB
Sanaa (sample patient)	SYN-0002	Female · 36 y	1990-07-22

TEST DATE	HEIGHT / WEIGHT	BMI	BLOOD PRESSURE
2026-10-05	162 cm · 58 kg	22.1	112/72 mmHg

DIET	ALLERGIES
Vegetarian · Indian	None

Abnormal results

12 of 54 outside reference range

Iron-deficiency anaemia

HGB 112 ↓ | MCV 76 ↓ | MCH 25.4 ↓ | RDW 17.8 ↑ | FERR 9 ↓ | IRON 7 ↓

Low haemoglobin with small, pale red cells (low MCV and MCH) and high RDW. Ferritin and serum iron are depleted, confirming low iron stores.

Reactive changes

PLT 410 ↑ | ESR 22 ↑

Platelets and ESR mildly raised, a pattern that often accompanies iron deficiency; CRP is within range.

Vitamin D deficiency

VIT-D 14 ↓ | PTH 74 ↑

Low 25-OH vitamin D with secondary rise in PTH; calcium remains in range.

Mild lipid elevation

CHOLT 4.6 ↑ | LDL 2.6 ↑

Total cholesterol and LDL slightly above the tight template ranges; HDL, triglycerides and ApoB are in range.

Metabolic & glucose

0 of 4 abnormal

CODE	TEST	RESULT	UNIT	REFERENCE	FLAG	RANGE
GLC	Glucose	**4.9**	mmol/L	3.5 - 5.4	–	
HGBA1C	Hemoglobin A1c	**5.3**	%	3.52 - 5.99	–	
C-PEPTIDE	Connecting peptide	**1.4**	ng/ml	0.8 - 3.5	–	
IGF-1	Insulin-like Growth Factor-1	**176**	ng/ml	115.0 - 307.0	–	

Lipids & cardiovascular risk

2 of 6 abnormal

CODE	TEST	RESULT	UNIT	REFERENCE	FLAG	RANGE
CHOLT	Total Cholesterol	**4.6**	mmol/L	3.0 - 4.0	H	
LDL	LDL Cholesterol	**2.6**	mmol/L	1.0 - 2.0	H	
HDL	HDL Cholestrol	**1.7**	mmol/L	1.0 - 2.7	–	
TRIG	Triglycerides	**0.9**	mmol/L	0.5 - 1.7	–	
APOB	Apolipoprotein B	**88**	mg/dl	40.0 - 125.0	–	
CRP	C-reactive protein	**1.4**	mg/l	1.0 - 3.0	–	

Liver & pancreas

0 of 12 abnormal

CODE	TEST	RESULT	UNIT	REFERENCE	FLAG	RANGE
ALT	Alanine Transaminase	**18**	U/L	5.0 - 45.0	–	
AST	Aspartate Transaminase	**20**	U/L	5.0 - 35.0	–	
ALP	Alkaline Phosphatase	**72**	U/L	30.0 - 110.0	–	
GGT	Gamma-GT	**14**	U/L	5.0 - 50.0	–	
BILIT	Total Bilirubin	**8**	umol/L	2.0 - 20.0	–	
BILID	Direct Bilirubin	**2.2**	umol/L	1.7 - 5.1	–	
ALB	Albumin	**41**	g/L	32.0 - 45.0	–	
GLOBT	Total Globulin	**29**	g/L	20.0 - 40.0	–	
PROT	Total Protein	**70**	g/L	60.0 - 80.0	–	
LDH	Lactate Dehydrogenase	**172**	u/l	125.0 - 220.0	–	
AAMY	Alpha-Amylase	**52**	U/L	25.0 - 120.0	–	
AFP	Alpha Fetoprotein	**1.8**	ng/ml	0.89 - 8.78	–	

Kidney & electrolytes

0 of 10 abnormal

CODE	TEST	RESULT	UNIT	REFERENCE	FLAG	RANGE
CREA	Creatinine	**64**	umol/L	60.0 - 110.0	–	
BUN	Blood Urea Nitrogen	**11.2**	mg/dl	9.8 - 20.1	–	
GFR	Glomerular Filtration Rate	**112**	ml/min/1.73m^2	90.0 - 120.0	–	
UA	Uric Acid	**0.26**	mmol/L	0.2 - 0.45	–	
NA+	Sodium	**139**	mmol/L	135.0 - 145.0	–	
K+	Potassium	**4.1**	mmol/L	3.5 - 5.0	–	
CL	Chloride	**99**	mmol/L	95.0 - 100.0	–	
CA	Calcium	**2.28**	mmol/L	2.15 - 2.64	–	

CODE	TEST	RESULT	UNIT	REFERENCE	FLAG	RANGE
P	Phosphorous	1.2	mmol/L	0.8 - 1.5	–	
MG+	Magnesium	1.8	mg/dl	1.6 - 2.6	–	

Bone & vitamin D

2 of 2 abnormal

CODE	TEST	RESULT	UNIT	REFERENCE	FLAG	RANGE
VIT-D	Vitamin D	14	ng/ml	20.0 - 50.0	L	
PTH	Parathyroid hormone	74	pg/ml	15.0 - 65.0	H	

Iron studies

2 of 2 abnormal

CODE	TEST	RESULT	UNIT	REFERENCE	FLAG	RANGE
IRON	Iron	7	umol/L	10.0 - 33.0	L	
FERR	Ferritin	9	ug/L	25.0 - 290.0	L	

Haematology

6 of 18 abnormal

CODE	TEST	RESULT	UNIT	REFERENCE	FLAG	RANGE
HGB	Hemoglobin	112	g/L	130.0 - 160.0	L	
HCT	Hematocrit	0.35	L/L	0.33 - 0.46	–	
RBC	Red Blood Cell	4.4	10^12/l	3.6 - 5.2	–	
MCV	Mean Corpuscular Volume	76	fL	80.0 - 100.0	L	
MCH	Mean Corpuscular Haemoglobin	25.4	pg	27.0 - 35.0	L	
MCHC	Mean Corpuscular Haemoglobin Concentration	318	g/L	300.0 - 350.0	–	
RDW	Red Cell Distribution Width	17.8	%	11.0 - 17.0	H	
PLT	Platelets	410	10^9/l	150.0 - 400.0	H	
MPV	Mean Platelet Volume	8.9	fl	6.0 - 12.0	–	
PDW	Platelet Distribution Width	12.4	fL	9.8 - 16.2	–	
WBC	White Blood Cell	6.4	x10^9/l	4.0 - 11.0	–	
NEUTR%	Neutrophils,%	56	%	40.0 - 80.0	–	
LYMPH%	Lymphocytes,%	34	%	20.0 - 40.0	–	
MONO%	Monocytes,%	6	%	2.0 - 10.0	–	
EOS%	Eosinophils,%	2.1	%	1.0 - 6.0	–	
BASO%	Basophils,%	0.5	%	0.0 - 2.0	–	
ATLYMPH	Atypical lymphocytes	1	%	0.0 - 12.0	–	

CODE	TEST	RESULT	UNIT	REFERENCE	FLAG	RANGE
ESR	Erythrocyte Sedimentation Rate	22	mm/hr	1.0 - 20.0	H	

All values are synthetic and were generated for demonstration of the Mulk Longevity report format. Reference ranges are those supplied in the source biomarker template and are not sex-specific. H = above range, L = below range. Range bar shows the reference interval as the shaded band; the marker is the result. Clinical comments are illustrative only.